MATS APPLICATION: TRANSIENT ESCAPE FROM A REFUSAL TRAJECTORY IN AN INSTRUCTION-TUNED TRANSFORMER

HIMANSHU SINGH
GITHUB REPOSITORY: HTTPS://GITHUB.COM/HIMANSHUVNM/TRANSIENT-REFUSAL-DYNAMICS
MY WEBSITE: HTTPS://HIMANSHUVNM.GITHUB.IO/
THE PENNSYLVANIA STATE UNIVERSITY

Executive Summary

Research Question. I investigated whether jailbreak outcomes in an instruction-tuned language model are associated with a localized change in internal representation dynamics. More specifically, I asked whether prompts that lead to non-refusal or provisionally successful jailbreak outcomes exhibit a measurable departure from an internal refusal-associated trajectory at particular transformer layers.

I chose this question because many mechanistic interpretability analyses treat model activations as static objects where one extracts a hidden state at a layer and asks whether it linearly separates two behaviors. Although useful, such analyses can confound behavior with prompt topic, wording, or style. I therefore tested a more dynamical hypothesis by asking whether the path of the residual-stream representation across layers contains a localized signature of a behavioral transition.

Initial Approach and Why I Changed It. I first tested a simple hidden-state classification setup using benign prompts and safety-related prompts. For Qwen2.5-3B-Instruct, I extracted the residual-stream representation of the final prompt token at each layer and constructed layerwise directions separating the two classes. The result initially appeared extremely strong with near-perfect AUROC across most layers. I did not consider this convincing. The benign and safety prompts differed substantially in semantic content and lexical cues, so the classifier could have been detecting concepts such as authorization, credentials, privacy, or security rather than anything specifically related to refusal behavior. Therefore, I constructed a synthetic set on the same topic containing benign, unsafe, and contextual-safety variants of the same underlying topic. Topic-held-out classification remained very strong. A further comparison between direct unsafe prompts and contextual safety prompts was also nearly perfectly separable. This indicated that prompt framing itself was strongly encoded and that static classification remained too easy to interpret mechanistically. This negative result changed the direction of the project. Instead of asking whether behaviors occupy different regions of the activation space, I asked whether the residual-stream trajectory undergoes a localized departure from a refusal-associated path.

Dynamical Formulation. For each prompt x, I extracted the final-prompt-token residual representation across the network, $h_0(x), h_1(x), \dots, h_L(x)$, where $h_\ell(x)$ denotes the hidden representation at layer ℓ. Because the residual-stream norm grows substantially with depth, I normalized each state,

$$\widehat{h}_\ell(x) = \frac{h_\ell(x)}{\|h_\ell(x)\|_2 + \varepsilon}.$$

I then constructed an independent refusal trajectory from a separate collection of matched unsafe prompts that the model consistently refused. Let r_ℓ denote the normalized refusal centroid at layer ℓ. For a new prompt x, define its distance from the refusal trajectory by

$$d_\ell(x) = \left\| \widehat{h}_\ell(x) - r_\ell \right\|_2 .$$

Then I measured the layerwise change in this distance using the following formula:

$$e_\ell(x) = d_{\ell+1}(x) - d_\ell(x).$$

Positive values of $e_\ell(x)$ indicate that the representation is moving away from the refusal trajectory at the transition from layer ℓ to $\ell + 1$. Exploratory analysis showed the most visible group separation in a late-layer region, approximately layers 22–30. Therefore, we define the escape score as follows:

$$E(x) \coloneqq \max_{\ell=22,\cdots,30} e_\ell(x).$$

Model, Data, and Experimental Setup. I used Qwen2.5-3B-Instruct and sampled 60 adversarial prompts from the JailbreakBench judge-comparison dataset. All model inference and hidden-state extraction were performed locally using `PyTorch-MPS` $-$ `M5`. For each prompt, I saved the generated response and the final-prompt-token residual representation across 37 hidden-state levels, each of dimension 2048. The behavioral outcome labels used for the final quantitative analysis were provisional. I combined an explicit-refusal detector with a semantic goal-response similarity measure and assigned each output to one of three categories, given as failed attack, successful attack, or ambiguous. Three ambiguous cases were excluded, leaving 57 examples - 22 provisional failures, 35 provisional successes.

Main Result. Overall, layer-to-layer representation movement was very similar between explicit refusals and nonrefusals. Thus, the main effect was not simply that one class exhibited larger hidden-state movement throughout the network. Instead, the more distinctive behavior appeared in the distance from the independent refusal trajectory. Provisionally successful cases showed a larger localized late-layer departure, particularly around layers 22–30. As shown in Figure 1, the two outcome groups exhibit broadly similar layer-to-layer dynamics across much of the network, but the provisionally successful cases show a larger localized departure from the independent refusal trajectory in the late-layer region, particularly across layers 22–30.

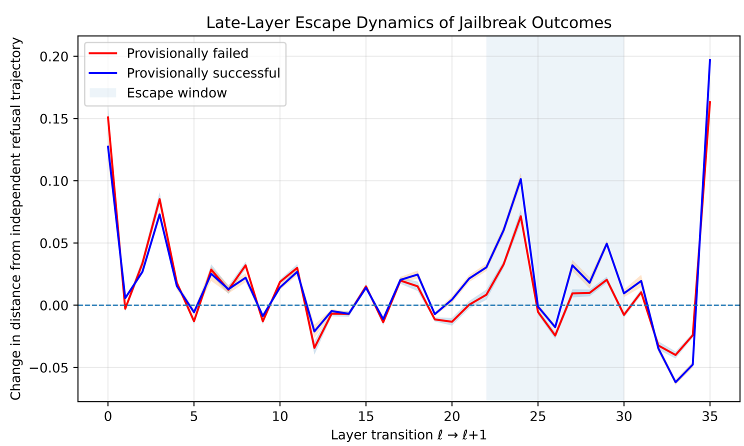

FIGURE 1. **Late-layer escape from an independently defined refusal trajectory.** Successful attacks exhibit a larger localized departure from the refusal trajectory in the late-layer region used to define the escape score, while overall layer-to-layer dynamics remain broadly similar across groups.

The mean escape scores were

$$\overline{E}_{\text{failure}} = 0.0715, \qquad \overline{E}_{\text{success}} = 0.1066.$$

Using $E(x)$ alone to predict the provisional outcome,

$$\text{AUROC} = 0.8325.$$

A 5000-trial label-permutation test gave $p \approx 2.0 \times 10^{-4}$, and the bootstrap 95% confidence interval for AUROC was $[0.693,\ 0.943]$. I ***interpret*** this as preliminary evidence that jailbreak-related behavior may be associated with a localized late-layer departure from a refusal-associated internal trajectory rather than with globally larger representational motion.

Strongest Evidence Against the Hypothesis. The strongest counterevidence came from the initial static classification experiments. They produced a nearly perfect separation, but follow-up controls showed that this result was strongly confounded by the semantic topic and prompt framing. This made it clear that high classification accuracy alone did not constitute mechanistic evidence. The second caution is that the final layer of the network exhibits a large representational jump for both successful and failed groups. Therefore, I do not interpret this final transition as jailbreak-specific. A third limitation is that the final behavioral labels are heuristic rather than produced by an independent established jailbreak judge. I attempted to use the JailbreakBench judge implementation, but package and environment compatibility prevented a clean evaluation within the available research window. Therefore, I consider the reported AUROC as exploratory rather than definitive.

Limitations and Next Steps. The present study uses a single model, a small sample of 60 benchmark prompts, and a layer window that emerged during the exploratory analysis rather than being preregistered. The behavioral labels are provisional and the analysis is correlational. I have not demonstrated that manipulating the identified late-layer region causally changes the refusal behavior. The most important next experiment is to reevaluate the same model responses using an independent jailbreak judge and to replicate the escape analysis on a second modern instruction-tuned model. If the late-layer effect persists, a causal intervention, such as activation steering or activation patching, could test whether suppressing or amplifying the identified escape changes the refusal behavior. The main lesson from this project is therefore not merely that a hidden-state statistic can predict jailbreak behavior. The more interesting observation is that static separation was too easy to trust, while a trajectory-based analysis produced a narrower and more falsifiable hypothesis about where refusal behavior may begin to diverge internally.

Email address, Singh: `himanshuvnm@gmail.com`

(Singh) Department of Meteorology and Atmospheric Science, College of Earth and Mineral Sciences, The Pennsylvania State University